Well-conditioned bases and a hybrid metric–parametric trial space for unsymmetric spectral elements

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Abstract

Unsymmetric (Petrov–Galerkin) finite elements with parametric test and metric trial functions reproduce polynomial displacement fields exactly on arbitrarily distorted meshes, including meshes with $\det \boldsymbol{J} \leq 0$. For spectral elements of degree p the metric trial functions are obtained from a generalised Vandermonde system whose conditioning and whose interpolation properties deteriorate rapidly with p and with the distortion. We separate the two effects and propose one remedy for each. (1) *Basis.* The trial space does not depend on the polynomial basis used to set up the interpolation problem, but the condition number $\kappa(\boldsymbol{V})$ does. A tensor-product Lagrange basis on the Gauss–Lobatto–Legendre points of the bounding box of the skew node coordinates makes $\boldsymbol{V}$ a permutation matrix on affine elements ($\kappa = 1$; metric and parametric functions coincide to round-off; the element matrix is symmetric to round-off for every p), and on distorted elements reduces $\kappa(\boldsymbol{V})$ to essentially the Lebesgue function of the interpolation at the reference nodes; compared with monomials, κ decreases by factors $4.6 \cdot 10^3$–$2.9 \cdot 10^4$ on distorted elements at $p = 16$, and the error of the computed shape functions by factors $3.6 \cdot 10^2$–$8.2 \cdot 10^3$. The Lebesgue constant itself is basis invariant and cannot be improved this way. (2) *Hybrid space.* Completeness requires only the complete polynomials $P_p(\boldsymbol{x})$ to be metric. The hybrid space $T_p = P_p(\boldsymbol{x}) \oplus \operatorname{span}\{L_a(\xi)L_b(\eta) : a + b > p\}$ keeps metric (Cartesian) polynomials for P_p and writes the remaining functions in parametric coordinates. It is complete, frame invariant, interpolatory, coincides with the conventional spectral element on bilinear elements, and its interpolant is the conventional one plus a polynomial correction; we derive a factorisation of its interpolation matrix that characterises unisolvence and a bound of its Lebesgue constant in terms of the Gauss–Lobatto–Legendre value. In numerical studies (single elements up to $p = 16$, Jacobian-unlimited patch tests, and a thick cylinder on meshes with fixed relative distortion, $p \leq 8$) the hybrid element keeps Lebesgue constants near the parametric level where those of Q_p in skew coordinates reach 10^4–10^8, reproduces polynomial fields to $\leq 10^{-9}$ also on folded meshes, has no eigenvalues with negative real part and only small imaginary parts ($\leq 1.8\,\%$ of the modulus), and converges with the optimal order p and quasi-optimally (discrete error 0.78–1.00 times the interpolation error) on all non-folded meshes, where the original unsymmetric element loses optimality beyond moderate distortion and the conventional element drops to order $\approx \lfloor p/2 \rfloor$. Limitations (folded elements at high p, the distortion dependence of the error constant) are documented.

1 Introduction

Isoparametric elements lose accuracy on distorted meshes because the parametric shape functions, composed with a non-affine element map, no longer span the complete polynomials in Cartesian coordinates [1, 11]. The unsymmetric finite element method of Rajendran and Liew [18] restores completeness by a Petrov–Galerkin formulation: the test functions remain parametric, the trial functions are *metric*, i.e. polynomials in physical coordinates that interpolate the nodal values. With the Jacobian determinant cancelled in the stiffness integral, the resulting elements reproduce every displacement field of total degree $\leq p$ exactly on arbitrarily distorted meshes, also when $\det \boldsymbol{J}$ changes sign (Jacobian-unlimited elements, conditions C1–C3 of Rajendran [17]). The approach has been developed for a wide range of element types and applications, e.g. [4, 5, 10, 12, 14–16, 19–21]; two known difficulties are the frame dependence of Cartesian metric functions, removed by skew (corotational) coordinates [13, 20], and the "interpolation failure" of the metric functions on strongly distorted elements [13, 17].

For high-order (spectral) elements, the metric trial functions of degree p in each coordinate are obtained element by element from a generalised Vandermonde system with $(p + 1)^2$ unknowns. Two different

properties of this construction degrade with p and with the element distortion, and they are often not distinguished:

1. the condition number $\kappa(\boldsymbol{V})$ of the interpolation matrix, which controls the round-off in the computed shape functions and depends on the polynomial basis used to set up the system;

2. the Lebesgue constant Λ of the interpolation in the trial space, which controls the approximation and the amplification of nodal perturbations; it is a property of the space and the node set and does not depend on the basis.

A companion study of the thick cylinder of Rajendran [17, Sec. 5.9] with $p = 2, \dots, 8$ (Section 5.5) shows the practical consequence: on meshes whose relative distortion does not decrease under refinement, the unsymmetric spectral element with metric functions Q_p in skew coordinates converges optimally only for moderate distortion; beyond, its convergence is erratic, and the loss correlates with Lebesgue constants of 10^3–10^8 rather than with the sign of $\det \boldsymbol{J}$.

This paper proposes one remedy for each property.

1. *A well-conditioned basis* for the metric interpolation problem (Section 3): a tensor-product Lagrange basis on the Gauss–Lobatto–Legendre (GLL) points of the bounding box of the skew node coordinates. We prove that the trial space and its Lebesgue constant are basis invariant, that the new basis makes $\boldsymbol{V}$ a permutation matrix on affine elements, and that on distorted elements $\kappa_1(\boldsymbol{V})$ equals the product of the reference Lebesgue function at the scaled nodes and the Lebesgue function of the metric interpolation at the reference points, i.e. the basis removes all ill-conditioning that is not inherent in the interpolation problem. It does *not* reduce the Lebesgue constant.

2. *A hybrid trial space* (Section 4) that uses metric polynomials only where completeness requires them, for $P_p(\boldsymbol{x})$, and parametric functions for the remaining $\frac{1}{2}p(p+1)$ functions. We establish completeness, frame invariance and the reduction to the conventional spectral element on bilinear elements, derive a factorisation of the interpolation matrix that characterises unisolvence, and show that the hybrid interpolant is the GLL interpolant plus the interpolation error of a polynomial, which bounds its Lebesgue constant by the GLL value plus a term that vanishes on bilinear elements.

Section 5 supports the claims by element-level studies up to $p = 16$, mesh-level studies (smooth fields, eigenvalues), Jacobian-unlimited patch tests on the meshes of Rajendran [17], and the thick-cylinder study with 1512 solutions. Section 6 summarises the advantages and the limitations.

2 Unsymmetric spectral elements

2.1 Petrov–Galerkin formulation

Let $\hat{\Omega} = [-1, 1]^2$ with the tensor grid of $(p+1)^2$ GLL points $\boldsymbol{\xi}_k$ and the Lagrange polynomials $N_k \in Q_p(\boldsymbol{\xi})$, $N_k(\boldsymbol{\xi}_l) = \delta_{kl}$. An element Ω^e is the image of the isoparametric map $\boldsymbol{x}(\boldsymbol{\xi}) = \sum_k N_k(\boldsymbol{\xi})\boldsymbol{x}_k$ with nodes $\boldsymbol{x}_k$; $\boldsymbol{J} = \partial\boldsymbol{x}/\partial\boldsymbol{\xi}$. For linear elasticity with elasticity matrix $\boldsymbol{D}$ and strain operator $\boldsymbol{\varepsilon}$, the unsymmetric element uses the parametric functions N_k as test functions and *metric* functions M_k as trial functions, which interpolate at the element nodes, $M_k(\boldsymbol{x}_l) = \delta_{kl}$, and span a space S^e. The element stiffness matrix is

$$\boldsymbol{K}^e = \int_{\hat{\Omega}} \left(\boldsymbol{B}_L^*\right)^{\mathsf{T}} \boldsymbol{D}\, \boldsymbol{B}_R \,\mathrm{d}\xi\,\mathrm{d}\eta, \qquad \boldsymbol{B}_L^* = \boldsymbol{B}_L \det \boldsymbol{J}, \tag{1}$$

where $\boldsymbol{B}_L^*$ is built from $\operatorname{adj}\boldsymbol{J}\,\nabla_{\boldsymbol{\xi}} N_k$ and $\boldsymbol{B}_R$ from $\nabla_{\boldsymbol{x}} M_k$ evaluated at $\boldsymbol{x}(\boldsymbol{\xi})$ [17]. The load vector is that of the conventional element. If $P_p(\boldsymbol{x}) \subset S^e$ on every element (completeness), the nodes are unisolvent for S^e, the metric functions are evaluated at the mapped quadrature points, and loads and the (polynomial) integrand of (1) are integrated exactly, every displacement field of total degree $\le p$ is reproduced exactly on any mesh, independently of the sign of $\det \boldsymbol{J}$ (distortion immunity). The conventional spectral element (PP) uses $S^e = \{N_k \circ \boldsymbol{x}^{-1}\}$ and $\boldsymbol{B}_L^{\mathsf{T}}\boldsymbol{D}\boldsymbol{B}_L \det \boldsymbol{J}$ as integrand.

The classical metric space is Q_p in the corotational skew coordinates $\hat{\boldsymbol{x}} = \boldsymbol{J}_0^{-1}(\boldsymbol{x} - \boldsymbol{x}_0)$ ($\boldsymbol{J}_0$, $\boldsymbol{x}_0$: Jacobian and image of the element centre), which are affine in $\boldsymbol{x}$ and frame invariant [20]; hence $P_p(\boldsymbol{x}) \subset Q_p(\hat{x}, \hat{y})$.

We denote the resulting element by PM. Given a basis $\{\phi_c\}_{c=1}^n$, $n=(p+1)^2$, of S^e, the metric functions at query points $\boldsymbol{x}_q$ follow from

$$\boldsymbol{V}\boldsymbol{M}=\boldsymbol{\Phi}, \qquad V_{ck}=\phi_c(\boldsymbol{x}_k), \quad \Phi_{cq}=\phi_c(\boldsymbol{x}_q), \quad M_{kq}=M_k(\boldsymbol{x}_q), \tag{2}$$

and analogously for the gradients.

2.2 Two measures of the metric interpolation

For nodal data $\boldsymbol{f}=(f_k)$ the metric interpolant is $\mathcal{I}_S\boldsymbol{f}=\sum_k f_k M_k$. Two quantities characterise this construction.

- The *condition number* $\kappa(\boldsymbol{V})=\|\boldsymbol{V}\|\,\|\boldsymbol{V}^{-1}\|$ of the linear system (2). With LU factorisation and partial pivoting the computed $\hat{\boldsymbol{M}}$ satisfies $\|\hat{\boldsymbol{M}}-\boldsymbol{M}\|/\|\boldsymbol{M}\| \lesssim c_n\,\kappa(\boldsymbol{V})\,u$ with unit round-off $u\approx 1.1\cdot 10^{-16}$ [9].
- The *Lebesgue constant*

$$\Lambda_S=\max_{\boldsymbol{x}\in\Omega^e}\lambda_S(\boldsymbol{x}), \qquad \lambda_S(\boldsymbol{x})=\sum_k |M_k(\boldsymbol{x})|, \tag{3}$$

  the norm of $\mathcal{I}_S$ from ℓ^∞ nodal data to $L^\infty(\Omega^e)$. It bounds the interpolation error, $\|u-\mathcal{I}_S u\|_\infty \le (1+\Lambda_S)\inf_{v\in S^e}\|u-v\|_\infty$ [3], and the amplification of perturbations of the nodal values (round-off, data errors) by the trial functions.

For the GLL tensor grid, Λ_{GLL} grows like $(\log p)^2$; it is 4.15 at $p=8$ and 6.0 at $p=16$. For the conventional element on any mesh $\Lambda_{\mathsf{PP}}=\Lambda_{\mathrm{GLL}}$, since $\lambda(\boldsymbol{x}(\boldsymbol{\xi}))=\sum_k|N_k(\boldsymbol{\xi})|$.

3 Improvement 1: a well-conditioned basis

3.1 Basis invariance

Proposition 1 (basis invariance)**.** *If $\boldsymbol{V}$ in (2) is non-singular, the metric functions are the unique $M_k\in S^e$ with $M_k(\boldsymbol{x}_l)=\delta_{kl}$; they, their Lebesgue function λ_S and the element matrix (1) do not depend on the basis $\{\phi_c\}$.*

Proof. For another basis $\boldsymbol{\psi}=\boldsymbol{C}\boldsymbol{\phi}$ with non-singular $\boldsymbol{C}$, $\boldsymbol{V}_\psi=\boldsymbol{C}\boldsymbol{V}_\phi$ and $\boldsymbol{\Phi}_\psi=\boldsymbol{C}\boldsymbol{\Phi}_\phi$, hence $\boldsymbol{V}_\psi^{-1}\boldsymbol{\Phi}_\psi=\boldsymbol{V}_\phi^{-1}\boldsymbol{\Phi}_\phi$; the same holds for the gradients. □

Completeness, frame invariance, distortion immunity and unisolvence ($\boldsymbol{V}$ non-singular for one basis iff for all) are therefore properties of the space S^e and the node set. The basis affects only the floating-point error, through $\kappa(\boldsymbol{V})$. Conversely, *no choice of basis can reduce* Λ_S: approximation properties and the amplification of nodal perturbations can only be improved by changing the space (Section 4).

3.2 Candidate bases and scaling

The original implementation uses monomials $\hat{x}^a\hat{y}^b$. For monomials on real nodes symmetric about the origin, the condition number grows at least exponentially with the degree [6]; on a parallelogram the 2D matrix is the Kronecker product of two 1D Vandermonde matrices, so $\kappa_2(\boldsymbol{V})$ is the square of the 1D value. We consider tensor-product bases $\phi_a(\tilde{s})\phi_b(\tilde{t})$ with Legendre polynomials, Chebyshev polynomials, or the Lagrange polynomials ℓ_a through the 1D GLL points. These polynomials are bounded on $[-1,1]$ but grow rapidly outside, while the skew coordinates of the nodes of a distorted element leave $[-1,1]^2$ (e.g. $|\hat{x}|\le 1.27$ for the trapezoid of Section 5.1). The node coordinates are therefore mapped affinely from their bounding box onto $[-1,1]^2$,

$$\tilde{s}=(\hat{x}-c_s)/r_s, \qquad \tilde{t}=(\hat{y}-c_t)/r_t, \tag{4}$$

with centre c and half-widths r of the box. The map is affine and computed from frame invariant quantities, so the space $Q_p(\hat{x},\hat{y})$ and frame invariance are preserved (Proposition 1).

Lemma 2 (affine elements)**.** *On a parallelogram, the Lagrange basis on the GLL points in the scaled coordinates (4) gives $\boldsymbol{V}=\boldsymbol{\Pi}$, a permutation matrix: $\kappa(\boldsymbol{V})=1$, $M_k=N_k\circ\boldsymbol{x}^{-1}$ exactly, and $\boldsymbol{K}^e$ of (1) equals the symmetric matrix of the conventional element.*

Proof. For an affine map $\boldsymbol{x} = \boldsymbol{x}_0 + \boldsymbol{J}_0\boldsymbol{\xi}$, $\hat{\boldsymbol{x}} = \boldsymbol{\xi}$. The bounding box of the nodes is $[-1,1]^2$, since the GLL grid contains the end points, so $(\tilde{s},\tilde{t}) = \boldsymbol{\xi}$ and $\phi_c(\boldsymbol{x}_k) = \ell_a(\xi_k)\ell_b(\eta_k) = \delta_{ck}$ up to the numbering. Then $\boldsymbol{M} = \boldsymbol{\Pi}^{\mathsf{T}}\boldsymbol{\Phi}$ is evaluated without arithmetic, $M_k(\boldsymbol{x}(\boldsymbol{\xi})) = N_k(\boldsymbol{\xi})$, and $\nabla_{\boldsymbol{x}} M_k = \boldsymbol{J}^{-\mathsf{T}}\nabla_{\boldsymbol{\xi}} N_k$, so $\boldsymbol{B}_L^* = \boldsymbol{B}_R \det \boldsymbol{J}$ and $\boldsymbol{K}^e = \int \boldsymbol{B}_R^{\mathsf{T}}\boldsymbol{D}\boldsymbol{B}_R \det \boldsymbol{J}\, \mathrm{d}\xi\, \mathrm{d}\eta$. □

Proposition 3 (distorted elements)**.** *Let $\hat{\boldsymbol{s}}_d$ be the reference GLL points in the scaled coordinates (4), so that $\phi_c(\hat{\boldsymbol{s}}_d) = \delta_{cd}$, let $\tilde{\boldsymbol{s}}_k$ be the scaled node coordinates, and let $\lambda_{\mathrm{ref}}(\boldsymbol{s}) = \sum_c |\phi_c(\boldsymbol{s})|$. Then*

$$(\boldsymbol{V}^{-1})_{kd} = M_k(\hat{\boldsymbol{s}}_d), \qquad \kappa_1(\boldsymbol{V}) = \max_k \lambda_{\mathrm{ref}}(\tilde{\boldsymbol{s}}_k)\ \max_d \lambda_S(\hat{\boldsymbol{s}}_d). \tag{5}$$

Proof. With the query points $\boldsymbol{x}_q = \hat{\boldsymbol{s}}_d$, $\boldsymbol{\Phi} = \boldsymbol{I}$ and $\boldsymbol{M} = \boldsymbol{V}^{-1}$. The 1-norm is the maximum absolute column sum: $\|\boldsymbol{V}\|_1 = \max_k \sum_c |\phi_c(\tilde{\boldsymbol{s}}_k)|$ and $\|\boldsymbol{V}^{-1}\|_1 = \max_d \sum_k |M_k(\hat{\boldsymbol{s}}_d)|$. □

Since $\tilde{\boldsymbol{s}}_k \in [-1,1]^2$, the first factor is bounded by the reference value Λ_{GLL}. With this basis, $\kappa(\boldsymbol{V})$ is therefore the Lebesgue function of the metric interpolation at the reference points of the bounding box, up to the factor Λ_{GLL}: the basis adds no ill-conditioning beyond that of the interpolation problem itself. Note that the points $\hat{\boldsymbol{s}}_d$ may lie outside Ω^e, where the metric polynomials extrapolate, so $\max_d \lambda_S(\hat{\boldsymbol{s}}_d)$ can exceed Λ_S.

The Legendre and Chebyshev bases have no such identity and are not exact on affine elements ($\kappa_2 = 55$ and 2.6 at $p = 16$, respectively), but behave similarly on distorted elements (Section 5.1). We therefore use the scaled Lagrange basis as the default of the metric space $Q_p(\hat{x},\hat{y})$.

4 Improvement 2: a hybrid metric–parametric trial space

4.1 Rationale and definition

Distortion immunity requires $P_p(\boldsymbol{x}) \subset S^e$; nothing else. The space $Q_p(\hat{x},\hat{y})$ satisfies this requirement with $\frac{1}{2}p(p+1)$ additional functions of total degree $p+1$ to $2p$ in physical coordinates. On a non-affine element the nodes $\boldsymbol{x}_k$, images of a tensor grid, do not form a tensor grid in $(\hat{x},\hat{y})$, and interpolation with these high-degree tensor terms on the perturbed grid drives Λ_S up; Proposition 7 below makes this precise, and Fig. 3 shows that removing these terms brings Λ back to the GLL level. The parametric functions, in contrast, interpolate on the exact GLL grid in $\boldsymbol{\xi}$. The hybrid space keeps metric functions only for P_p and uses parametric functions for the remainder.

Definition 4 (hybrid trial space)**.** With the Legendre polynomials L_a,

$$C_p = \mathrm{span}\{L_a(\xi)L_b(\eta) :\ 0 \le a,b \le p,\ a+b > p\}, \qquad \tilde{T}_p^e = \{\, q \circ \boldsymbol{x} + r :\ q \in P_p(\hat{x},\hat{y}),\ r \in C_p \,\}. \tag{6}$$

The trial functions are the $M_k \in \tilde{T}_p^e$ with $M_k(\boldsymbol{\xi}_l) = \delta_{kl}$, evaluated through $\boldsymbol{\xi}$: $u_h(\boldsymbol{x}(\boldsymbol{\xi})) = \sum_k u_k M_k(\boldsymbol{\xi})$.

C_p is the $L^2(\hat{\Omega})$-orthogonal complement of $P_p(\boldsymbol{\xi})$ in $Q_p(\boldsymbol{\xi})$, so $Q_p(\boldsymbol{\xi}) = P_p(\boldsymbol{\xi}) \oplus C_p$; this fixes the complement canonically. (Another complement C' of $P_p(\boldsymbol{\xi})$ in $Q_p(\boldsymbol{\xi})$, e.g. $\mathrm{span}\{\xi^a\eta^b : a+b > p\}$, gives the same space only if $C' \subset P_p(\boldsymbol{x}) \circ \boldsymbol{x} \oplus C_p$, which holds when $P_p(\boldsymbol{\xi}) \subset \tilde{T}_p^e$, e.g. on bilinear elements, but not in general.) In the implementation P_p is represented by $L_a(\tilde{s})L_b(\tilde{t})$, $a+b \le p$, in the scaled skew coordinates (4), and the interpolation matrix is

$$\boldsymbol{V}_T = \begin{bmatrix} \boldsymbol{Q} \\ \boldsymbol{R} \end{bmatrix}, \qquad Q_{jk} = q_j(\boldsymbol{x}_k), \quad R_{ik} = r_i(\boldsymbol{\xi}_k), \tag{7}$$

with the $d = \frac{1}{2}(p+1)(p+2)$ basis functions q_j of P_p and the $n-d$ functions r_i of C_p. Gradients: $\nabla_{\boldsymbol{x}}(q \circ \boldsymbol{x}) = \nabla_{\boldsymbol{x}} q$ exactly, $\nabla_{\boldsymbol{x}} r = \boldsymbol{J}^{-\mathsf{T}}\nabla_{\boldsymbol{\xi}} r$. We denote the element with (1) and (6) by PH.

4.2 Properties

Proposition 5. *(a) $P_p(\boldsymbol{x}) \circ \boldsymbol{x} \subset \tilde{T}_p^e$ for every element map: PH is complete and distortion immune for fields of total degree $\le p$, independently of the sign of $\det \boldsymbol{J}$.*

(b) $\tilde{T}_p^e$ is frame invariant and contains the rigid-body modes.

(c) If $\boldsymbol{x}(\boldsymbol{\xi})$ is bilinear (straight-sided element) and $\boldsymbol{V}_T$ is non-singular, then $\tilde{T}_p^e = Q_p(\boldsymbol{\xi})$, $M_k = N_k$, and PH *is the conventional (symmetric) spectral element. In particular, on affine elements* PH*,* PM *and* PP *coincide.*

(d) The trial functions are interpolatory, so the GLL nodal-quadrature mass matrix is diagonal.

Proof. (a) by definition. (b) $P_p(\hat{x},\hat{y}) = P_p(\boldsymbol{x})$ is invariant under rigid motions, and so is $\boldsymbol{\xi}$; $P_1(\boldsymbol{x})$ contains the rigid-body modes. (c) For bilinear $\boldsymbol{x}$, $x, y \in Q_1(\boldsymbol{\xi})$, so a monomial of total degree $j \le p$ in (x, y) is a product of j functions in $Q_1(\boldsymbol{\xi})$ and lies in $Q_j(\boldsymbol{\xi}) \subset Q_p(\boldsymbol{\xi})$ [1]; $C_p \subset Q_p(\boldsymbol{\xi})$. Hence $\tilde{T}_p^e \subset Q_p(\boldsymbol{\xi})$, with equality of dimensions if $\boldsymbol{V}_T$ is non-singular. (d) $M_k(\boldsymbol{\xi}_l) = \delta_{kl}$. □

By (c), PH differs from the conventional element only on elements with curved edges (or non-bilinear interior node placement); it is unsymmetric only there. A consequence of the same argument is that the conventional Lagrange spectral element already contains $P_p(\boldsymbol{x})$ on straight-sided quadrilaterals, so its loss of accuracy under distortion stems from curved (non-bilinear) element maps, such as those of the mid-side node distortions studied in Section 5.

Unisolvence. Let $\mathcal{I}_{\mathrm{ref}}$ denote interpolation in $Q_p(\boldsymbol{\xi})$ at the GLL grid, and Π_{low} the projection onto the Legendre coefficients with $a + b \le p$.

Proposition 6 (factorisation)**.** *Let $\boldsymbol{V}_{\mathrm{ref}}$ be the Legendre Vandermonde matrix of the GLL grid (rows $L_a(\xi_k)L_b(\eta_k)$, all $a, b \le p$) and $\boldsymbol{A} \in \mathbb{R}^{d\times d}$ the matrix of the map $q \mapsto \Pi_{\mathrm{low}}\mathcal{I}_{\mathrm{ref}}(q \circ \boldsymbol{x})$ on P_p, A_{ij} = coefficient of the i-th low-order Legendre product in $\mathcal{I}_{\mathrm{ref}}(q_j \circ \boldsymbol{x})$. Then*

$$|\det \boldsymbol{V}_T| = |\det \boldsymbol{V}_{\mathrm{ref}}|\,|\det \boldsymbol{A}|, \tag{8}$$

so the nodes are unisolvent for $\tilde{T}_p^e$ iff $\boldsymbol{A}$ is non-singular.

Proof. Order the rows of $\boldsymbol{V}_{\mathrm{ref}}$ as low-order ($a + b \le p$) rows $\boldsymbol{L}$ followed by the rows $\boldsymbol{R}$ of (7). Each row of $\boldsymbol{Q}$ is a vector of nodal values and has a unique expansion in the rows of $\boldsymbol{V}_{\mathrm{ref}}$: $\boldsymbol{Q} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{L} + \boldsymbol{B}\boldsymbol{R}$, the coefficients being those of $\mathcal{I}_{\mathrm{ref}}(q_j \circ \boldsymbol{x})$. Hence $\boldsymbol{V}_T = \begin{bmatrix} \boldsymbol{A}^{\mathsf{T}} & \boldsymbol{B} \\ \boldsymbol{0} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{L} \\ \boldsymbol{R} \end{bmatrix}$. □

On an affine element $q \circ \boldsymbol{x} \in P_p(\boldsymbol{\xi})$, $\mathcal{I}_{\mathrm{ref}}$ is exact, Π_{low} is the identity on $P_p(\boldsymbol{\xi})$, and $\boldsymbol{A}$ is the matrix of the invertible composition with an affine map. Since $\det \boldsymbol{A}$ depends continuously on the nodes, the nodes are unisolvent for every map sufficiently close to an affine one. For strongly non-affine maps unisolvence is not guaranteed; Section 5.5 shows that it fails on some folded elements at $p \ge 7$, exactly where $\boldsymbol{A}$ becomes singular. Since $\boldsymbol{V}_{\mathrm{ref}}$ is fixed and well conditioned, $\mathrm{rcond}(\boldsymbol{A})$ is a cheap element-level indicator ($d \times d$ instead of $n \times n$).

Structure of the interpolant and Lebesgue constant.

Proposition 7. *If $\boldsymbol{A}$ is non-singular, the hybrid interpolant of nodal data $\boldsymbol{f}$ is*

$$\mathcal{I}_T\boldsymbol{f} = \mathcal{I}_{\mathrm{ref}}\boldsymbol{f} + (\mathrm{Id} - \mathcal{I}_{\mathrm{ref}})(q_{\boldsymbol{f}} \circ \boldsymbol{x}), \qquad q_{\boldsymbol{f}} = \sum_j c_j q_j, \quad \boldsymbol{A}\boldsymbol{c} = \Pi_{\mathrm{low}}\mathcal{I}_{\mathrm{ref}}\boldsymbol{f}, \tag{9}$$

i.e. $M_k = N_k + (\mathrm{Id} - \mathcal{I}_{\mathrm{ref}})(q_k \circ \boldsymbol{x})$ with $q_k = q_{\boldsymbol{e}_k}$. Consequently

$$\Lambda_T \le \Lambda_{\mathrm{GLL}} + \|\boldsymbol{E}_{\boldsymbol{x}}\|\,\|\boldsymbol{A}^{-1}\|\,\gamma_p, \qquad \boldsymbol{E}_{\boldsymbol{x}} : \boldsymbol{c} \mapsto (\mathrm{Id} - \mathcal{I}_{\mathrm{ref}})\Big(\sum_j c_j q_j \circ \boldsymbol{x}\Big), \tag{10}$$

with $\gamma_p = \|\Pi_{\mathrm{low}}\mathcal{I}_{\mathrm{ref}}\|_{\ell^\infty\to\ell^\infty}$ depending on p only and the norms $\ell^\infty \to L^\infty(\hat{\Omega})$ for $\boldsymbol{E}_{\boldsymbol{x}}$. On bilinear elements $\boldsymbol{E}_{\boldsymbol{x}} = \boldsymbol{0}$ and $\Lambda_T = \Lambda_{\mathrm{GLL}}$.

Proof. Write $\mathcal{I}_T\boldsymbol{f} = q \circ \boldsymbol{x} + r$. Interpolation at the GLL grid is exact on $Q_p(\boldsymbol{\xi}) \ni r$, so $\mathcal{I}_{\mathrm{ref}}\boldsymbol{f} = \mathcal{I}_{\mathrm{ref}}(q \circ \boldsymbol{x}) + r$, i.e. $r = \mathcal{I}_{\mathrm{ref}}\boldsymbol{f} - \mathcal{I}_{\mathrm{ref}}(q \circ \boldsymbol{x})$, which lies in C_p iff $\Pi_{\mathrm{low}}\mathcal{I}_{\mathrm{ref}}(q \circ \boldsymbol{x}) = \Pi_{\mathrm{low}}\mathcal{I}_{\mathrm{ref}}\boldsymbol{f}$, i.e. $\boldsymbol{A}\boldsymbol{c} = \Pi_{\mathrm{low}}\mathcal{I}_{\mathrm{ref}}\boldsymbol{f}$. Substituting r gives (9); (10) follows from the triangle inequality and $\|\mathcal{I}_{\mathrm{ref}}\| = \Lambda_{\mathrm{GLL}}$. For bilinear maps $q \circ \boldsymbol{x} \in Q_p(\boldsymbol{\xi})$ (Proposition 5(c)), so $(\mathrm{Id} - \mathcal{I}_{\mathrm{ref}})(q \circ \boldsymbol{x}) = 0$. □

Proposition 7 states that the hybrid element is the conventional spectral element plus a correction that restores completeness, and that this correction is an *interpolation error* of polynomials composed with the element map. It is small whenever the map is close to bilinear: for $\boldsymbol{x} = \boldsymbol{x}_{\mathrm{bil}} + \delta\, \boldsymbol{g}$ with a smooth perturbation $\boldsymbol{g}$, $\|\boldsymbol{E}_{\boldsymbol{x}}\| = O(\delta)$, since $q \circ \boldsymbol{x}_{\mathrm{bil}} \in Q_p(\boldsymbol{\xi})$. For curved elements with smooth edges $q \circ \boldsymbol{x}$ is a polynomial of degree p times the degree of the map, and $(\mathrm{Id} - \mathcal{I}_{\mathrm{ref}})(q \circ \boldsymbol{x})$ decays with the smoothness of the map. The same argument applies to $Q_p(\hat{x}, \hat{y})$, with $q \in Q_p(\hat{x}, \hat{y})$, Π_{low} replaced by the identity and an $n \times n$ matrix $\boldsymbol{A}_Q$; but then the correction contains the interpolation errors of $\hat{x}^a \hat{y}^b$ up to total degree $2p$, which do not vanish even for bilinear maps ($\hat{x}^p \hat{y}^p \circ \boldsymbol{x} \in Q_{2p}(\boldsymbol{\xi}) \setminus Q_p(\boldsymbol{\xi})$ in general). This is consistent with Fig. 3: on the straight-sided trapezoid $\Lambda_T = \Lambda_{\mathrm{GLL}}$, whereas Λ_Q grows to $4 \cdot 10^7$ at $p = 16$.

Element matrix. In (1) the columns of $\boldsymbol{B}_R$ that belong to P_p are polynomial in $\boldsymbol{\xi}$, so the Jacobian cancellation is retained for them; the columns of the complement contain $\boldsymbol{J}^{-1}$, and the corresponding integrand is $\propto (\mathrm{adj}\, \boldsymbol{J}^{\mathsf{T}} \nabla_{\boldsymbol{\xi}} N)^{\mathsf{T}} \boldsymbol{D} (\mathrm{adj}\, \boldsymbol{J}^{\mathsf{T}} \nabla_{\boldsymbol{\xi}} r) / \det \boldsymbol{J}$ (in strain-matrix form), rational as for curved isoparametric elements. Polynomial fields are unaffected: for $\boldsymbol{u} \in P_p(\boldsymbol{x})$ the interpolant is exact, $\sum_k u_k \nabla M_k = \nabla \boldsymbol{u}$ is polynomial, and the integrand $(\boldsymbol{B}_L^*)^{\mathsf{T}} \boldsymbol{D} \boldsymbol{\varepsilon}(\boldsymbol{u})$ is the same as for PM. On folded elements the complement part has a singular integrand where $\det \boldsymbol{J} = 0$ inside the element; PH is undefined if $\det \boldsymbol{J} = 0$ at a quadrature point. The cost per element equals that of PM plus one evaluation of $\boldsymbol{J}^{-1}$ per point; the parametric coordinates of the quadrature points are known in all element routines.

5 Numerical studies

All computations use the USEM-MATLAB code base (GNU Octave 8.4, IEEE double precision), in which the scaled Lagrange basis is the default of the metric space $Q_p(\hat{x}, \hat{y})$ (PM) and the hybrid space is available as a second trial space (PH); PP is the conventional spectral element on the same nodes. GLL nodes are used throughout. The scripts and data are listed in Section 6.

5.1 Test elements and meshes

Single elements: a parallelogram (affine), a trapezoid with straight edges (bilinear map), an element with two curved edges ("curved") and one with angular and curved-edge distortion ("general"), $p = 1, \ldots, 16$. *Mesh level:* a 3×2 mesh on $[0, 3] \times [0, 2]$ with curved-edge distortion 0.15, 0.20 and combined angular and curved-edge distortion 0.30 (minimum $\det \boldsymbol{J} = 0.16$, 0.09, 0.014). *Patch tests:* the square panel of Rajendran [17, Secs. 5.5–5.7], 2×2 elements whose internal mid-side nodes are rotated by $\theta = 90°$–$270°$ (every element has $\det \boldsymbol{J} \le 0$ somewhere), with interior nodes from the bilinear map of the corners. *Convergence:* the thick cylinder of Rajendran [17, Sec. 5.9] (quarter annulus $1 \le r \le 3$, plane stress, $E = 1000$, $\nu = 0.25$, internal pressure 1000), $2N \times 2N$ elements, $N = 1, 2, 4, 8$, generated with gmsh [7]; in each 2×2 super-element the internal mid-side nodes are rotated by $\alpha \in \{0°, 15°, 30°, 45°, 60°, 90°\}$, so that the *relative* distortion is the same on all meshes and the element maps are not asymptotically affine (Fig. 1). Folded elements occur for $\alpha = 45°$ ($N \ge 2$) and $\alpha \ge 60°$. The arcs are represented exactly (Gordon–Hall blending [8]). Three fields are computed: the Lamé solution, a manufactured polynomial of total degree 5, and the analytic field $u_x = \frac{1}{10}(\sin 3X \cos 2Y + e^{XY})$, $u_y = \frac{1}{10}(\cos 2X \sin 3Y + Y e^{-X})$, $X = x/3$, $Y = y/3$, with body forces and tractions derived from the exact fields. Errors: relative energy norm e_E (strains of the respective trial functions, signed $\det \boldsymbol{J}$) and relative maximum nodal error e_n. Cases in which the code base finds the interpolation matrix singular to working precision (rcond $< \varepsilon$) are marked "rej.".

5.2 Element level: conditioning and Lebesgue constants

Figure 2 shows $\kappa_2(\boldsymbol{V})$ and the error of the computed shape functions for worst-case (unit) nodal data, $e_\delta = \max_{k,l} |M_k(\boldsymbol{x}_l) - \delta_{kl}|$, for the space $Q_p(\hat{x}, \hat{y})$ in four bases. With monomials κ_2 grows by a factor ≈ 5.7 per degree already on the parallelogram ($3.5 \cdot 10^{11}$ at $p = 16$), close to $(1 + \sqrt{2})^2$, i.e. $(1 + \sqrt{2})^p$ for each 1D factor [6]; the scaled Lagrange basis gives $\kappa_2 = 1$ for every p (Lemma 2), and the element matrix is symmetric to round-off, whereas its asymmetry grows to $2 \cdot 10^{-6}$ with monomials (Table 1). On the distorted elements the three alternatives lie within a factor 3 of one another and are better than monomials by factors $4.6 \cdot 10^3$–$2.9 \cdot 10^4$ at $p = 16$; e_δ follows $\kappa_2 u$ and decreases by factors $3.6 \cdot 10^2$–$8.2 \cdot 10^3$ at $p = 16$ (e.g. trapezoid: $3.0 \cdot 10^{-2} \to 7.1 \cdot 10^{-6}$). For $p \le 6$ all bases give $e_\delta < 5 \cdot 10^{-13}$. Without the bounding-box

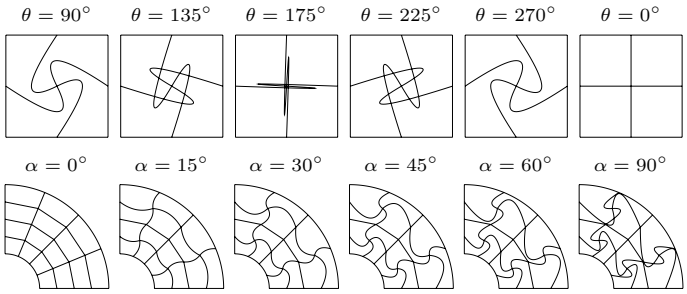


Figure 1: Top: panel meshes of the Jacobian-unlimited patch test (internal mid-side nodes rotated by θ). Bottom: thick-cylinder meshes for $N = 2$ (4×4 elements); the finer meshes repeat the pattern on N^2 super-elements.

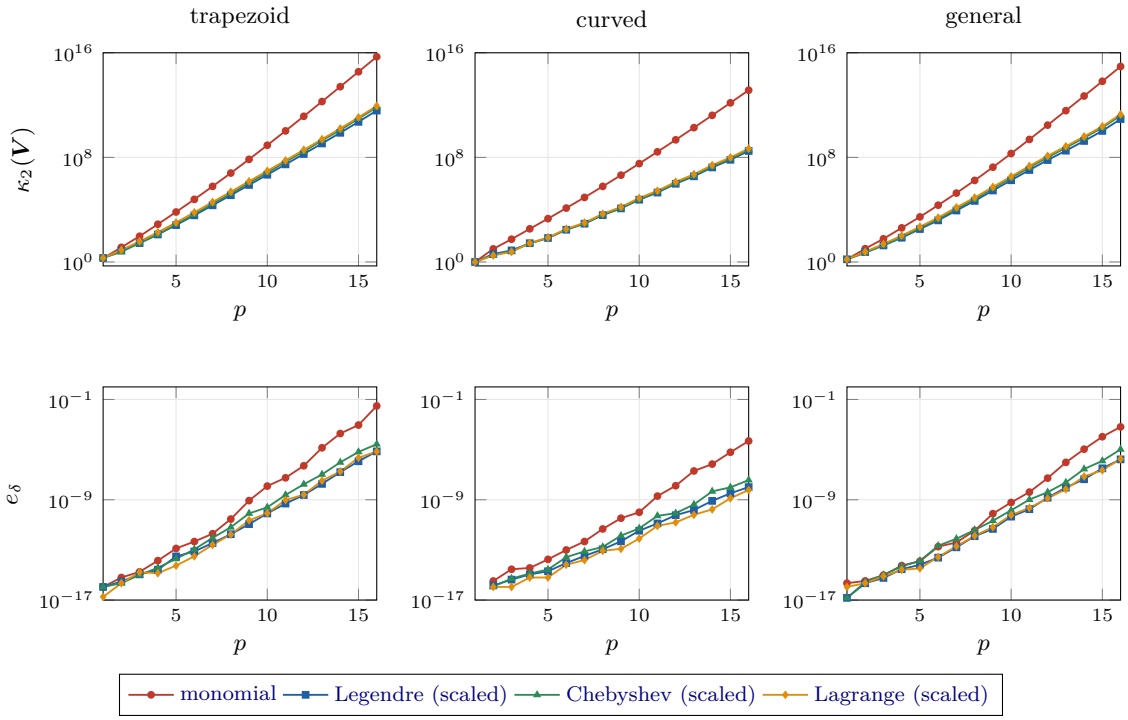


Figure 2: Space $Q_p(\hat{x}, \hat{y})$ on the distorted test elements: condition number of the interpolation matrix (top) and error of the Kronecker property of the computed shape functions (bottom) for four bases.

scaling (4) the orthogonal and Lagrange bases lose more than three orders of magnitude on the trapezoid ($\kappa_2 = 2.2 \cdot 10^{15}$ instead of $8.6 \cdot 10^{11}$ at $p = 16$).

As Proposition 1 predicts, the reproduction of polynomials and the Lebesgue constant are the same in all bases. Figure 3 compares Λ of $Q_p(\hat{x}, \hat{y})$ and of the hybrid space T_p on the three distorted elements. For $Q_p(\hat{x}, \hat{y})$, Λ grows exponentially, to $4.0 \cdot 10^7$ (trapezoid), $5.5 \cdot 10^4$ (curved) and $6.7 \cdot 10^6$ (general) at $p = 16$. For T_p it equals Λ_{GLL} on the trapezoid (bilinear map, Proposition 7) and stays within 2 % (curved) and 40 % (general) of it (6.1 and 8.5 at $p = 16$).

5.3 Mesh level: smooth field and eigenvalues

On the distorted 3×2 meshes (Fig. 4) the patch test of degree p (harmonic polynomial field) fails for the conventional element (10^{-7}–10^{-2}) and is passed by both unsymmetric elements, with errors growing to 10^{-13}–10^{-12} at $p = 10$ for $Q_p(\hat{x}, \hat{y})$ and $\leq 3.1 \cdot 10^{-15}$ for T_p. For the field $\boldsymbol{u} = (e^x \cos y, -e^x \sin y)$ all three elements converge exponentially; for $p \leq 8$, $Q_p(\hat{x}, \hat{y})$ is in most cases the more accurate of the two unsymmetric elements (by factors up to 17; exceptions: curved 0.20, $p = 6, 8$; combined, $p = 7$), at $p = 9, 10$ T_p is more accurate or within a factor 1.5, and T_p is 3.6–150 times more accurate than PP for $p = 5, \ldots, 10$ (curved 0.20). The decisive difference is in the spectrum of the generalised eigenvalue problem $\boldsymbol{K}\boldsymbol{\phi} = \lambda \boldsymbol{M}\boldsymbol{\phi}$ (consistent mass, homogeneous Dirichlet conditions; reference: conventional element on the undistorted

Table 1: Parallelogram (affine element), $Q_p(\hat{x},\hat{y})$: asymmetry $\|\boldsymbol{K}^e - \boldsymbol{K}^{e\mathsf{T}}\|_F/\|\boldsymbol{K}^e\|_F$ of the element matrix (zero in exact arithmetic) and condition number of $\boldsymbol{V}$.

p	4	6	8	10	12	16
monomial	$2.2\cdot10^{-15}$	$8.4\cdot10^{-14}$	$4.2\cdot10^{-12}$	$1.3\cdot10^{-10}$	$2.9\cdot10^{-9}$	$2.0\cdot10^{-6}$
Lagrange (scaled)	$1.6\cdot10^{-15}$	$2.0\cdot10^{-15}$	$4.6\cdot10^{-15}$	$1.5\cdot10^{-14}$	$1.5\cdot10^{-14}$	$3.4\cdot10^{-14}$
$\kappa_2(\boldsymbol{V})$, monomial	$2.9\cdot10^{2}$	$9.2\cdot10^{3}$	$2.9\cdot10^{5}$	$9.6\cdot10^{6}$	$3.2\cdot10^{8}$	$3.5\cdot10^{11}$
$\kappa_2(\boldsymbol{V})$, Lagrange (scaled)	1.0	1.0	1.0	1.0	1.0	1.0

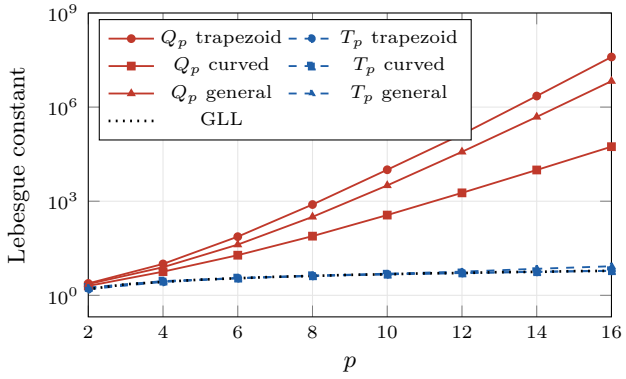


Figure 3: Lebesgue constant of the metric interpolation in $Q_p(\hat{x},\hat{y})$ and in the hybrid space T_p on the distorted test elements; GLL: parametric interpolation (any element).

mesh, $p = 12$). With $Q_p(\hat{x},\hat{y})$, from $p = 5$–7 on 2–25 eigenvalues have negative real part, the maximum of $|\operatorname{Im}\lambda|/|\lambda|$ approaches 1, and the error of the eight lowest eigenvalues stagnates at 10^{-5}–10^{-3} or reaches $O(1)$ (curved 0.20, $p = 6$: 1.9). With T_p no eigenvalue has negative real part, $|\operatorname{Im}\lambda|/|\lambda| \le 1.8\cdot10^{-2}$ over the whole spectrum, and the errors of the eight lowest eigenvalues do not exceed those of the conventional element by more than 6 % and are smaller in most cases ($4.5\cdot10^{-7}$ for both at $p = 10$). An eigenvalue with negative real part is an exponentially growing mode for every time integrator; for transient and modal analyses at $p \gtrsim 5$ on curved meshes this disqualifies $Q_p(\hat{x},\hat{y})$.

5.4 Jacobian-unlimited patch test

On the panel meshes the shear load case of Rajendran [17] has a cubic exact solution, contained in both unsymmetric spaces for $p \ge 3$. Table 2 gives the maximum nodal error over the five distorted meshes (Fig. 1). Both unsymmetric elements reproduce the field on all meshes with folded and overlapping elements; the round-off level of PM reaches $1.1\cdot10^{-5}$ at $p = 8$, that of PH $6.7\cdot10^{-8}$. The conventional element has errors of 13–140 %. For $p = 2$ (9-node element, corner-centroid interior node, 5×5 Gauss points) PH passes the constant and linear strain patch tests on all meshes and converges with the energy-norm rates 2.02 and 2.21 on the folded mesh sequences B and C of the paper (PM-QUAD9*: 2.00, 1.98; conventional element: 0.70, 2.50, the latter from a negative displacement on the coarsest mesh).

5.5 Thick cylinder: accuracy, convergence and stability

Polynomial field. Table 3 gives the maximum nodal error for the degree-5 field over all meshes and $p = 5,\dots,8$. Both unsymmetric elements are exact in exact arithmetic; in floating point the error of PM grows with the distortion to $1.1\cdot10^{-4}$ (45°) and $1.6\cdot10^{-2}$ (folded mesh 90°), that of PH stays $\le 9.9\cdot10^{-10}$ on every mesh on which it can be built. With exact integration (PM: $n_q \ge p^2 + p/2$ points) the residual of the exact nodal values is $9.9\cdot10^{-7}$ for PM and $5.0\cdot10^{-11}$ for PH ($\alpha = 60°$, $N = 2$, $p = 8$), with $\sigma_{\min}/\sigma_{\max}(\boldsymbol{K}) = 1.1\cdot10^{-15}$, $4.8\cdot10^{-8}$ and $2.9\cdot10^{-6}$ for PM, PH and PP.

Convergence. Figure 5 shows the energy error of the analytic field and of the Lamé field at $\alpha = 30°$, Table 4 the rates on the finest pair of meshes. On these non-asymptotically-affine meshes the conventional

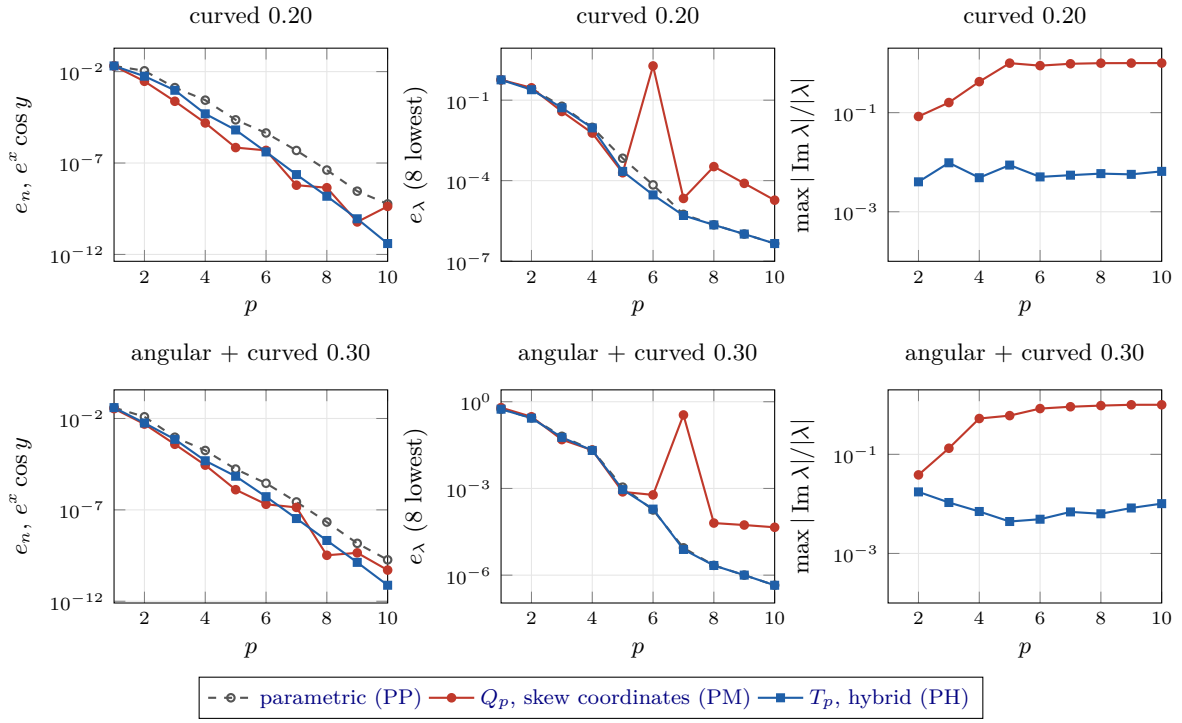


Figure 4: 3×2 meshes with curved-edge distortion: nodal error for the field $(e^x \cos y, -e^x \sin y)$ (left), error of the real parts of the eight eigenvalues of smallest modulus (centre), and largest relative imaginary part over the whole spectrum (right; zero for the symmetric conventional element).

Table 2: Jacobian-unlimited patch test (cubic field), maximum relative nodal error over the panel meshes with $\theta = 90°, \ldots, 270°$.

	$p=3$	$p=4$	$p=5$	$p=6$	$p=7$	$p=8$
parametric (PP)	$1.6 \cdot 10^{-1}$	$4.9 \cdot 10^{-1}$	$1.5 \cdot 10^{-1}$	$2.2 \cdot 10^{-1}$	$1.3 \cdot 10^{-1}$	1.4
$Q_p(\hat{x}, \hat{y})$ (PM)	$1.3 \cdot 10^{-13}$	$3.3 \cdot 10^{-11}$	$9.8 \cdot 10^{-11}$	$6.5 \cdot 10^{-11}$	$1.1 \cdot 10^{-9}$	$1.1 \cdot 10^{-5}$
T_p (PH)	$6.4 \cdot 10^{-13}$	$1.2 \cdot 10^{-11}$	$1.6 \cdot 10^{-11}$	$1.5 \cdot 10^{-10}$	$2.7 \cdot 10^{-10}$	$6.7 \cdot 10^{-8}$

element converges with $\approx \lfloor p/2 \rfloor$, because the parametric space contains $P_k(\boldsymbol{x})$ only for $2k \le p$ [1]. PM is optimal for $\alpha = 15°$, $p \le 6$, but at $\alpha = 30°$ only for $p \le 3$; beyond, its error is erratic and can increase under refinement (analytic field, $p = 4$: $e_E = 2.2 \cdot 10^{-4}$ at $N = 4$, $4.3 \cdot 10^{-4}$ at $N = 8$). PH converges with the optimal order p for all p at 15° and 30° (Lamé: 1.96–7.87). At $N = 8$, $\alpha = 30°$ (analytic field), its energy error is 8–$2.6 \cdot 10^4$ times smaller than that of PP, and 1.4–$6.7 \cdot 10^4$ times smaller than that of PM for $p \ge 3$ (equal at $p = 2$). Under p-refinement (Fig. 6) PH converges exponentially to $3.5 \cdot 10^{-13}$ (30°) and $1.1 \cdot 10^{-11}$ (45°, folded elements), PM only for $\alpha \le 15°$.

Approximation versus stability. Inserting the exact nodal values into the trial functions gives the interpolation error of each space; the ratio of discretisation to interpolation error isolates the stability of the discrete problem (Table 5). For PH it lies between 0.78 and 1.00 in all 158 cases on the meshes without folds with errors above the round-off level, for PP between 0.84 and 1.00 (165 cases): both are quasi-optimal with a constant close to one. For PM it ranges from 0.07 to 10^5; at $\alpha = 30°$, $p \ge 3$ the discrete PM solution is up to 64 times worse *or* up to five times better than its interpolant, i.e. it does not inherit the approximation property of its space. The PM space itself also approximates worse where its Lebesgue constant is large: at $\alpha = 30°$, $p \le 6$, the PM interpolant of the analytic field is up to nine times less accurate than the PH interpolant.

Lebesgue constants, conditioning and spectra. Table 6 lists the maxima over the elements of the finest mesh. The Lebesgue constant of PM reaches 10^3 at 30°, $p = 4$ and $1.3 \cdot 10^8$ at 45°, $p = 8$; that of

Table 3: Thick cylinder, polynomial field (degree 5): maximum relative nodal error over $N = 1, \dots, 8$ and $p = 5, \dots, 8$; in parentheses the number of rejected cases (singular interpolation matrix).

	$\alpha = 0°$	$\alpha = 15°$	$\alpha = 30°$	$\alpha = 45°$	$\alpha = 60°$	$\alpha = 90°$
PM	$1.6 \cdot 10^{-11}$	$3.8 \cdot 10^{-10}$	$7.6 \cdot 10^{-9}$	$1.1 \cdot 10^{-4}$	$1.2 \cdot 10^{-5}$ (4)	$1.6 \cdot 10^{-2}$ (3)
PH	$4.3 \cdot 10^{-13}$	$3.3 \cdot 10^{-13}$	$5.1 \cdot 10^{-13}$	$9.9 \cdot 10^{-10}$	$2.8 \cdot 10^{-10}$ (3)	$9.7 \cdot 10^{-10}$ (3)

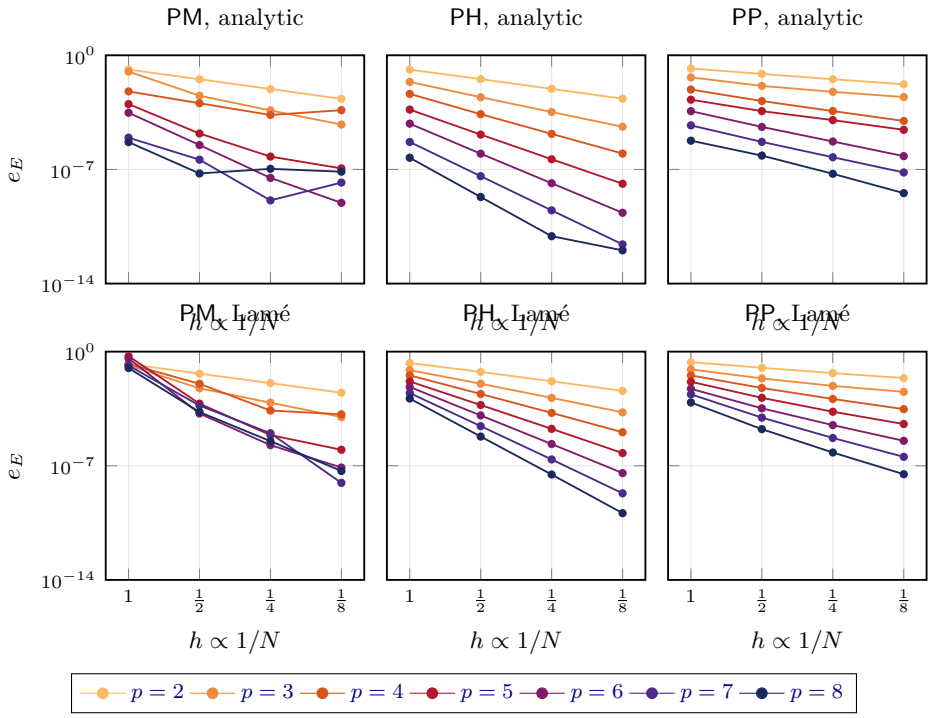


Figure 5: Thick cylinder, $\alpha = 30°$ (no folded elements): energy error versus element size for the analytic field (top) and the Lamé field (bottom).

PH stays within 30 % of the GLL value up to 15° and ≤ 95 at 30°. $\kappa_2(\boldsymbol{V})$ of PM exceeds its Lebesgue constant by up to two orders of magnitude (extrapolation at the reference points, Proposition 3); that of PH is 10^2–10^5 times smaller than that of PM at 30°, $p \geq 6$. At mesh level, the smallest singular values of the global stiffness matrices of the three elements agree to 1 % (14 cases, $\alpha \leq 30°$), but the largest singular value of PM exceeds that of PP by up to a factor 344 (30°, $N = 4$, $p = 6$), whereas that of PH stays within 1.2 of it. In the expansion of the error of the analytic field in the singular triplets of $\boldsymbol{K}$, a single one of the six lowest triplets carries 21–68 % of the PM error in four of five cases with $\alpha \geq 30°$; for PH and PP the error is spread over the spectrum (largest single share 2–8 %), and the lowest 50 singular values of PH agree with those of PP to 2.3 %.

Limits of the hybrid space. (i) On the folded meshes with $\alpha \geq 60°$ the interpolation matrix of PH becomes singular in 6 of the 24 cases with $p \geq 6$ (that of PM in 7). Table 7 confirms Proposition 6: rcond($\boldsymbol{A}$) follows rcond($\boldsymbol{V}_T$) within a factor 30, the determinant identity (8) holds to $\leq 3 \cdot 10^{-13}$ (relative) for rcond($\boldsymbol{V}_T$) $\geq 5 \cdot 10^{-6}$ and degrades with the conditioning as expected for computed determinants, and the singular cases are exactly those with rcond($\boldsymbol{A}$) $\leq 4 \cdot 10^{-16}$ (all other cases: $\geq 2 \cdot 10^{-15}$). (ii) On the same meshes the PH error for $p = 2$ (60°) and $p \leq 4$ (90°) is non-monotone in N and of order 10^{-2}–1, as for the other two elements; the singular complement integrand (Section 4) is one possible cause. (iii) The error constant of PH depends on the distortion: for the Lamé field at $\alpha = 15°$, where PM is stable, PH is up to 4.1 times less accurate than PM ($p \leq 7$), because its interpolation error is 1.2–5.7 times larger ($N = 8$); the parametric complement approximates the r^{-2} terms of this field worse than the full Q_p in skew coordinates.

Table 4: Thick cylinder: observed rates of e_E between $N = 4$ and $N = 8$ (optimal: p); "–": error at round-off level.

		$p=2$	$p=3$	$p=4$	$p=5$	$p=6$	$p=7$	$p=8$
analytic, $\alpha = 15°$	PM	2.00	2.99	3.99	4.99	5.98	–	–
	PH	2.00	3.00	4.00	4.99	6.01	–	–
	PP	1.08	1.23	2.21	2.03	2.94	3.39	4.05
analytic, $\alpha = 30°$	PM	2.00	2.86	-0.98	2.40	–	–	–
	PH	2.00	3.00	4.01	5.00	6.02	–	–
	PP	1.02	1.06	2.01	1.98	2.98	3.11	3.94
Lamé, $\alpha = 15°$	PM	1.97	2.91	3.90	4.81	5.83	–	–
	PH	1.96	2.94	3.94	4.92	5.91	6.88	7.87
	PP	1.07	1.43	2.16	2.81	3.49	4.19	4.78
Lamé, $\alpha = 30°$	PM	1.97	2.94	0.76	2.99	4.56	–	–
	PH	1.96	2.96	3.94	4.94	5.92	6.90	7.87
	PP	1.03	1.21	2.07	2.51	3.20	3.85	4.42

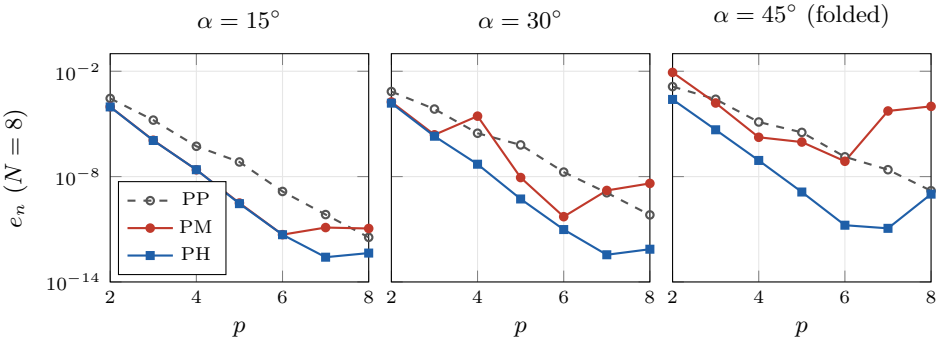


Figure 6: Thick cylinder, analytic field: p-convergence of the nodal error on the finest mesh.

6 Discussion and conclusions

The metric trial functions of unsymmetric spectral elements suffer from two distinct deficiencies that grow with the polynomial degree and the element distortion: an ill-conditioned interpolation *system* and an ill-conditioned interpolation *problem*. The first is a matter of the basis, the second of the trial space.

Basis. The trial space, its Lebesgue constant, distortion immunity and the exact element matrix are basis invariant (Proposition 1). The scaled tensor-product Lagrange basis on GLL points makes the interpolation matrix a permutation on affine elements, so the unsymmetric element coincides with the conventional one to round-off there (symmetric $\boldsymbol{K}^e$ for all p), and on distorted elements it reduces $\kappa(\boldsymbol{V})$ to the Lebesgue function of the interpolation at the reference points (Proposition 3). Compared with monomials it lowers κ by $4.6 \cdot 10^3$–$2.9 \cdot 10^4$ and the error of the computed shape functions by $3.6 \cdot 10^2$–$8.2 \cdot 10^3$ on the distorted test elements at $p = 16$, and it removes the spurious rejection of elements by a singularity check (five element–degree combinations on the panel meshes). Its benefit is confined to round-off: approximation, the amplification of nodal perturbations and the stability of the discrete problem are unchanged, and the static results of the thick cylinder change only at round-off level. A better basis *cannot* improve the Lebesgue constant.

Hybrid space. Completeness requires metric functions only for $P_p(\boldsymbol{x})$. The hybrid space T_p (Definition 4) keeps them for P_p and represents the remaining $\frac{1}{2}p(p+1)$ functions in parametric coordinates. It retains distortion immunity, frame invariance and interpolation, reduces to the conventional spectral element on straight-sided elements, and its interpolant is the conventional one plus the interpolation error of a polynomial composed with the element map (Proposition 7), which bounds its Lebesgue constant by the GLL value plus a term that vanishes for bilinear maps. Unisolvence is equivalent to the non-singularity of a $d \times d$ matrix $\boldsymbol{A}$ (Proposition 6). Numerically, the hybrid space

Table 5: Thick cylinder, $N = 8$: energy error of the discrete solution divided by the energy error of the interpolant of the exact field in the same trial space.

		$p = 2$	$p = 3$	$p = 4$	$p = 5$	$p = 6$	$p = 7$	$p = 8$
analytic, $\alpha = 15°$	PM	0.99	0.98	1.00	0.98	1.01	–	–
	PH	0.99	0.97	0.98	0.96	0.98	–	–
	PP	0.97	0.94	0.96	0.93	0.96	0.95	0.96
analytic, $\alpha = 30°$	PM	0.91	0.20	64.38	3.40	0.38	–	–
	PH	0.96	0.93	0.93	0.89	0.90	–	–
	PP	0.94	0.92	0.92	0.89	0.93	0.93	0.93
Lamé, $\alpha = 15°$	PM	1.00	0.99	0.99	0.98	0.99	–	–
	PH	0.98	0.95	0.96	0.95	0.96	0.94	0.95
	PP	0.98	0.96	0.98	0.97	0.98	0.97	0.98
Lamé, $\alpha = 30°$	PM	0.99	0.48	14.98	5.46	8.75	–	–
	PH	0.93	0.91	0.92	0.89	0.90	0.88	0.89
	PP	0.95	0.93	0.96	0.93	0.95	0.94	0.95

Table 6: Thick cylinder, $N = 8$: Lebesgue constant Λ (16×16 sample points per element, a lower bound) and condition number $\kappa_2(\boldsymbol{V})$ of the interpolation matrix, maximum over the elements.

		$p = 2$	$p = 3$	$p = 4$	$p = 5$	$p = 6$	$p = 7$	$p = 8$
Λ, $\alpha = 15°$	PM	2.5	6.0	$1.5 \cdot 10^1$	$3.7 \cdot 10^1$	$1.3 \cdot 10^2$	$1.2 \cdot 10^4$	$1.3 \cdot 10^4$
	PH	1.8	2.4	2.8	3.1	3.5	4.5	4.2
Λ, $\alpha = 30°$	PM	$1.0 \cdot 10^1$	$2.4 \cdot 10^2$	$1.1 \cdot 10^3$	$1.6 \cdot 10^3$	$6.2 \cdot 10^3$	$4.2 \cdot 10^6$	$1.2 \cdot 10^6$
	PH	2.0	4.8	7.5	$1.6 \cdot 10^1$	$2.5 \cdot 10^1$	$7.4 \cdot 10^1$	$9.5 \cdot 10^1$
Λ, $\alpha = 45°$	PM	$1.9 \cdot 10^2$	$1.5 \cdot 10^3$	$1.3 \cdot 10^4$	$5.0 \cdot 10^5$	$8.3 \cdot 10^5$	$1.4 \cdot 10^6$	$1.3 \cdot 10^8$
	PH	2.8	$1.0 \cdot 10^1$	$2.1 \cdot 10^1$	$4.7 \cdot 10^1$	$9.1 \cdot 10^1$	$2.6 \cdot 10^2$	$5.5 \cdot 10^2$
	GLL	1.56	2.22	2.68	3.00	3.41	3.47	4.15
$\kappa_2(\boldsymbol{V})$, $\alpha = 15°$	PM	4.8	$1.2 \cdot 10^1$	$6.0 \cdot 10^1$	$2.9 \cdot 10^2$	$1.9 \cdot 10^3$	$3.1 \cdot 10^5$	$2.4 \cdot 10^5$
	PH	5.5	9.7	$1.9 \cdot 10^1$	$4.0 \cdot 10^1$	$9.4 \cdot 10^1$	$2.3 \cdot 10^2$	$6.0 \cdot 10^2$
$\kappa_2(\boldsymbol{V})$, $\alpha = 30°$	PM	$3.8 \cdot 10^1$	$1.7 \cdot 10^3$	$5.9 \cdot 10^3$	$8.5 \cdot 10^3$	$8.6 \cdot 10^4$	$2.6 \cdot 10^8$	$2.6 \cdot 10^8$
	PH	9.0	$2.4 \cdot 10^1$	$6.0 \cdot 10^1$	$1.8 \cdot 10^2$	$7.0 \cdot 10^2$	$2.7 \cdot 10^3$	$9.8 \cdot 10^3$
$\kappa_2(\boldsymbol{V})$, $\alpha = 45°$	PM	$8.4 \cdot 10^2$	$1.5 \cdot 10^4$	$2.1 \cdot 10^5$	$2.9 \cdot 10^7$	$2.8 \cdot 10^8$	$4.7 \cdot 10^9$	$7.9 \cdot 10^{11}$
	PH	$1.8 \cdot 10^1$	$9.1 \cdot 10^1$	$4.1 \cdot 10^2$	$2.4 \cdot 10^3$	$3.3 \cdot 10^4$	$3.2 \cdot 10^5$	$4.0 \cdot 10^6$

- keeps the Lebesgue constant at or near the GLL value on single elements up to $p = 16$ (≤ 8.5, against $5.5 \cdot 10^4$–$4.0 \cdot 10^7$ for Q_p in skew coordinates) and ≤ 95 on the thick-cylinder meshes without folds (against up to $4.2 \cdot 10^6$);
- reproduces polynomial fields to $\leq 10^{-9}$ on all meshes, including folded ones, where Q_p in skew coordinates is limited to 10^{-5}–10^{-2} by round-off;
- produces no eigenvalues with negative real part and imaginary parts $\leq 1.8\,\%$ of the modulus, where Q_p in skew coordinates produces up to 25 unstable modes on curved meshes;
- converges with the optimal order p and quasi-optimally (discrete error 0.78–1.00 times the interpolation error) on all non-folded meshes of a mesh sequence with fixed relative distortion, on which Q_p in skew coordinates is optimal only for moderate distortion and the conventional element converges with $\approx \lfloor p/2 \rfloor$; the gain over the conventional element reaches $2.6 \cdot 10^4$ in the energy norm ($p = 7$, $\alpha = 30°$, $N = 8$).

Limitations. (i) No stability theory (discrete inf–sup condition [2]) is available for either unsymmetric element; quasi-optimality of PH is observed, not proven. (ii) Unisolvence of T_p is guaranteed only near affine maps; it fails on some folded elements at $p \geq 7$, and the complement part of the element matrix has a singular integrand where $\det \boldsymbol{J}$ changes sign; on folded meshes PH is reliable only for $p \geq 4$–5. (iii) The error constant of PH depends on the distortion: where Q_p in skew coordinates is stable, it can approximate

Table 7: Unisolvence of the hybrid space on the thick-cylinder meshes: minimum over the elements of rcond of $\boldsymbol{V}_T$, of $\boldsymbol{A}$ (Proposition 6) and of $\boldsymbol{V}$ of $Q_p(\hat{x}, \hat{y})$; number of elements with $\mathrm{rcond}(\boldsymbol{V}_T) < \varepsilon$.

α	N	p	$\min \mathrm{rcond}(\boldsymbol{V}_T)$	$\min \mathrm{rcond}(\boldsymbol{A})$	$\min \mathrm{rcond}(\boldsymbol{V}_Q)$	singular
0°	8	8	$1.9 \cdot 10^{-3}$	$5.5 \cdot 10^{-2}$	$1.0 \cdot 10^{-2}$	0
15°	8	8	$7.5 \cdot 10^{-5}$	$5.0 \cdot 10^{-4}$	$4.0 \cdot 10^{-7}$	0
30°	8	6	$1.0 \cdot 10^{-4}$	$5.5 \cdot 10^{-4}$	$2.9 \cdot 10^{-6}$	0
30°	8	7	$2.1 \cdot 10^{-5}$	$1.1 \cdot 10^{-4}$	$6.3 \cdot 10^{-10}$	0
30°	8	8	$5.3 \cdot 10^{-6}$	$2.7 \cdot 10^{-5}$	$6.3 \cdot 10^{-10}$	0
45°	8	6	$2.8 \cdot 10^{-6}$	$1.3 \cdot 10^{-5}$	$1.2 \cdot 10^{-9}$	0
45°	8	8	$1.8 \cdot 10^{-8}$	$8.1 \cdot 10^{-8}$	$4.7 \cdot 10^{-13}$	0
60°	4	8	$1.4 \cdot 10^{-17}$	$9.8 \cdot 10^{-17}$	$7.5 \cdot 10^{-19}$	4
60°	8	6	$3.1 \cdot 10^{-16}$	$2.2 \cdot 10^{-15}$	$1.3 \cdot 10^{-16}$	0
60°	8	7	$5.2 \cdot 10^{-19}$	$6.5 \cdot 10^{-18}$	$1.7 \cdot 10^{-18}$	8
60°	8	8	$3.4 \cdot 10^{-20}$	$3.1 \cdot 10^{-19}$	$6.1 \cdot 10^{-21}$	8
90°	1	7	$7.8 \cdot 10^{-16}$	$7.8 \cdot 10^{-15}$	$1.5 \cdot 10^{-15}$	0
90°	1	8	$6.1 \cdot 10^{-18}$	$6.1 \cdot 10^{-17}$	$5.3 \cdot 10^{-19}$	1
90°	8	6	$1.2 \cdot 10^{-14}$	$9.3 \cdot 10^{-14}$	$6.0 \cdot 10^{-15}$	0
90°	8	7	$3.9 \cdot 10^{-17}$	$3.9 \cdot 10^{-16}$	$2.8 \cdot 10^{-17}$	8
90°	8	8	$3.9 \cdot 10^{-20}$	$9.4 \cdot 10^{-19}$	$1.2 \cdot 10^{-19}$	16

better (by factors up to 5.7 for the Lamé field at $\alpha = 15°$, up to 17 for $e^x \cos y$ on curved meshes). (iv) The studies are two-dimensional, linear-elastic and restricted to $p \leq 16$ (element level) and $p \leq 8$ (convergence); 3D elements and non-linear problems remain to be examined.

For high-order unsymmetric spectral elements we therefore recommend the scaled Lagrange basis for any metric space, and the hybrid space T_p whenever $p \gtrsim 4$, the distortion is not small, or the spectrum matters (transient and modal analysis).

Reproduction

Code base: USEM-MATLAB after commit `e293da1` (`ShapeFunctions2D_Metric`, `p.metricBasis`, `p.metricMethod`). Element and mesh level: `studies/metric_basis/*.m`; patch tests: `studies/rajendran_panel`; thick cylinder: `studies/thick_cylinder`; unisolvence: `studies/manuscript_metric/matlab/unisolvence_check.m`; tables and data of this paper: `studies/manuscript_metric/make_data.py`.

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